Chapter 7: Exponential Functions

MULTIPLE CHOICE

1. Which set of properties does the function $y = 2^x$ have?

A no x-intercept, no y-intercept
B x-intercept is 1, no y-intercept
C no x-intercept, y-intercept is 1
D x-intercept is 0, y-intercept is 0

ANS: C DIF: Easy OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: intercepts | exponential function

2. Which choice best describes the function $y = 6^x$?

A both increasing and decreasing
B decreasing
C increasing
D neither increasing nor decreasing

ANS: C DIF: Easy OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions KEY: increasing | decreasing

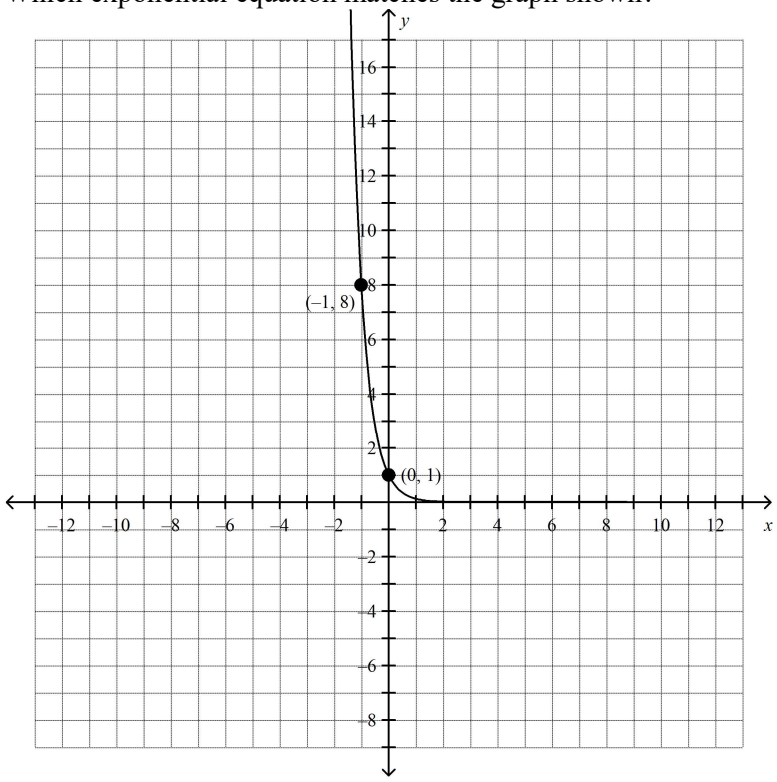
3. Which set of properties is correct for the function $y = \left(\frac{1}{9}\right)^x$?

A domain $\{x | x \in \mathrm{R}\}$, range $\{y | y > 0, y \in \mathrm{R}\}$
B domain $\{x | x \in \mathrm{R}\}$, range $\{y | y \geq 0, y \in \mathrm{R}\}$
C domain $\{x | x \in \mathrm{R}\}$, range $\{y | y \leq 0, y \in \mathrm{R}\}$
D domain $\{x | x \in \mathrm{R}\}$, range $\{y | y < 0, y \in \mathrm{R}\}$

ANS: A DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions KEY: domain | range

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4. Which exponential equation matches the graph shown?



A $y = \left(\frac{1}{8}\right)^x$

B $y = 8^x$

C $y = -\left(\frac{1}{8}\right)^x$

D $y = -8^x$

ANS: A DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: equation | graph | exponential function

5. A radioactive sample with an initial mass of 1 mg has a half-life of 9 days. What is the equation that models the exponential decay, A, for time, t, in 9-day intervals?

A $A = \left(\frac{1}{2}\right)^t$

B $A = \left(\frac{1}{2}\right)^{\frac{9}{t}}$

C $A = 2^{9t}$

D $A = 2^{\frac{9}{t}}$

ANS: A DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: modelling | exponential decay

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6. A colony of ants has an initial population of 750 and triples every day. Which function can be used to model the ant population, p, after t days?

A $p(t) = 3(750)^t$ **C** $p(t) = 750\left(\frac{1}{3}\right)^t$

B $p(t) = \frac{1}{3}(750)^t$ **D** $p(t) = 750(3)^t$

ANS: D DIF: Easy OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: modelling | exponential growth

7. A bacteria colony initially has 1500 cells and doubles every week. Which function can be used to model the population, p, of the colony after t days?

A $p(t) = 1500(3)^t$ **C** $p(t) = 1500(2)^{\frac{t}{7}}$

B $p(t) = 1500(2)^t$ **D** $p(t) = 1500(3)^{\frac{t}{7}}$

ANS: C DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: modelling | exponential growth

8. An investment of \$150 is placed into an account that earns interest, compounded annually, at a rate of 5% for 12 years. The amount, A, in the account can be modelled by the function $A = 150(1.05)^t$, where t is the time, in years. What is the domain of this function?

A $\{t | 0 < t < 12, t \in \mathrm{R}\}$ **C** $\{t | t > 12, t \in \mathrm{R}\}$

B $\{t | 0 \le t \le 12, t \in \mathrm{R}\}$ **D** $\{t | t \ge 12, t \in \mathrm{R}\}$

ANS: B DIF: Easy OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: range | domain | exponential function

9. A rumour is being spread in a community that the mayor is going to retire before the next election. Initially, 6 people were told the rumour and each of these people told 6 others the following day. Each of the 6 people told by the initial people told 6 others the day after they were told. If this process keeps up, how many people will be told the rumour on the eighth day?

A 1 679 616 people **C** 10 077 696 people

B 279 936 people **D** 48 people

ANS: A DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions KEY: exponential growth

10. To the nearest year, how long would an investment need to be left in the bank at 5%, compounded annually, for the investment to triple?

A 15 years **C** 28 years

B 26 years **D** 23 years

ANS: D DIF: Easy OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: compound interest

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11. Jennifer deposited some money into an account that pays 7% per year, compounded annually. Today her balance is \$300. How much was in the account 10 years ago, to the nearest cent?

[Hint: Use $P = A(1+i)^{-n}$.]

A \$163.18
B \$30.00
C \$42.86
D \$152.50

ANS: D DIF: Average OBJ: Section 7.2 NAT: RF10
TOP: Transformations of Exponential Functions
KEY: modelling | exponential decay

12. Mohamed purchased a car for \$16 000. It depreciates by 20% of its current value every year. How much will the car be worth 8 years after it is purchased?

A \$80 000
B \$2000
C \$2684.35
D \$68 797.07

ANS: C DIF: Average OBJ: Section 7.2 NAT: RF10
TOP: Transformations of Exponential Functions
KEY: modelling | exponential decay

13. The equation $A = 90\left(\frac{1}{5}\right)^n$ can also be written as

A $A = 90(5)^n$
B $A = 45(5)^{-n}$
C $A = 45(5)^n$
D $A = 90(5)^{-n}$

ANS: D DIF: Easy OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: exponential function | negative exponents

14. For the exponential function $y = 100\left(\frac{1}{9}\right)^x$, which of the following statements is *not* true?

A The graph of the function is increasing.
B The graph of the function is decreasing.
C The domain is the set of real numbers.
D The range is the set of real numbers greater than zero.

ANS: A DIF: Easy OBJ: Section 7.1 | Section 7.2
NAT: RF9
TOP: Characteristics of Exponential Functions | Transformations of Exponential Functions
KEY: increasing | decreasing | domain | range

15. Which of the following transformations maps the function $y = 8^x$ onto the function $y = 8^{x+5} + 7$?

A a horizontal shift of 5 units to the left and a vertical shift of 7 units up
B a horizontal shift of 5 units to the right and a vertical shift of 7 units down
C a horizontal shift of 5 units to the right and a vertical shift of 7 units up
D a horizontal shift of 5 units to the left and a vertical shift of 7 units down

ANS: A DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

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16. Which function results when the graph of $y = 6^x$ is translated 2 units down?

A $y = 6^{x-2}$

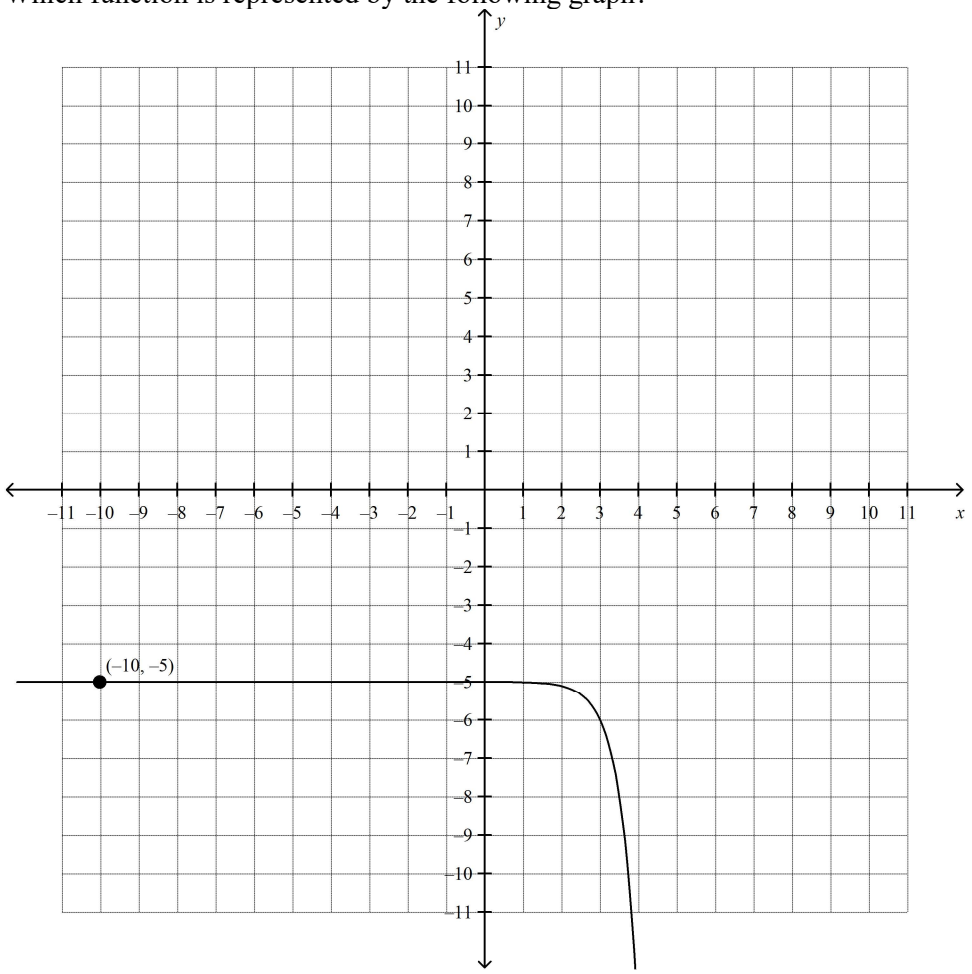
B $y = 6^{x+2}$

C $y = 6^x - 2$

D $y = 6^x + 2$

ANS: C DIF: Easy OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

17. Which function is represented by the following graph?



A $y = -9(9)^{(x+4)} + 5$

B $y = 9(9)^{-(x+4)} + 5$

C $y = -9(9)^{(x-4)} - 5$

D $y = 9(9)^{-(x-4)} - 5$

ANS: C DIF: Difficult OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

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18. Which function results when the graph of the function $y = 4^x$ is reflected in the *y*-axis, compressed vertically by a factor of $\frac{1}{5}$, and shifted 2 units down?

A $y = \frac{1}{5}(4)^{-x} - 2$

B $y = \frac{1}{5}(4)^{x} - 2$

C $y = \frac{1}{5}(4)^{x} + 2$

D $y = \frac{1}{5}(4)^{-x} + 2$

ANS: A DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

19. What is the exponential equation for the function that results from the transformations listed being applied to the base function $y = 9^x$?
- a reflection in the *y*-axis
- a vertical stretch by a factor of 6
- a horizontal stretch by a factor of 7

A $y = -7(9)^{\frac{x}{6}}$

B $y = 6(9)^{\frac{-x}{7}}$

C $y = 7(9)^{\frac{x}{6}}$

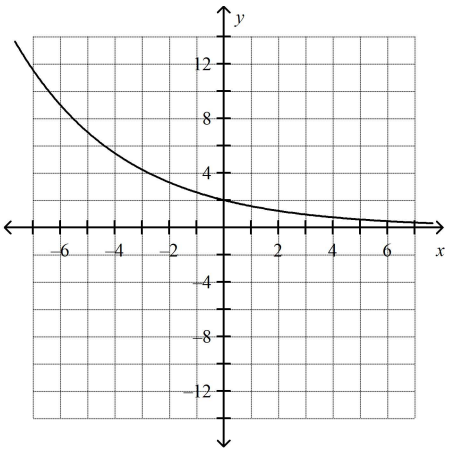
D $y = -6(9)^{\frac{x}{7}}$

ANS: B DIF: Easy OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

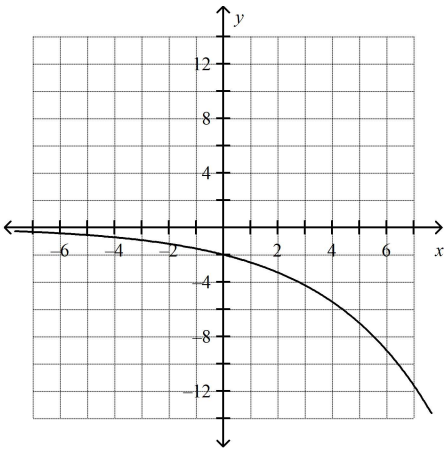
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20. Which graph represents the function $y = 2\left(\frac{7}{9}\right)^x$?

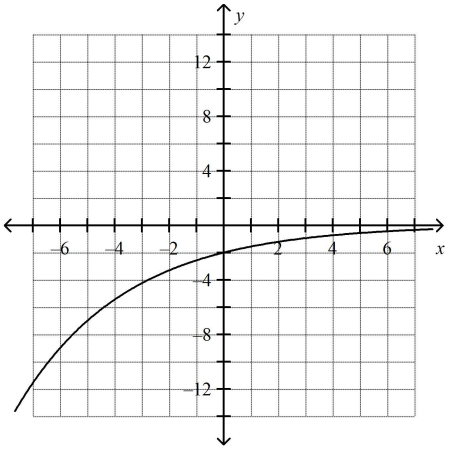
A



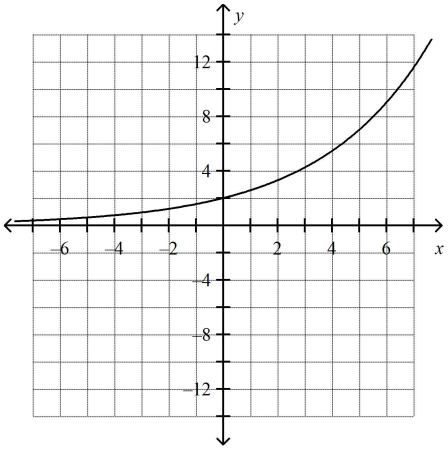
C



B



D



ANS: A DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: graph | transformations of exponential functions

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21. Which equation can be used to model the given information, where the population has been rounded to the nearest whole number?

Year (x)	Population (y)
0	100
1	104
2	108
3	112
4	117
5	122

A $y = 100(1.04)^x$
B $y = 100(1.4)^x$
C $y = 100(1.04)^{x-1}$
D $y = 100(1.4)^{x-1}$

ANS: A DIF: Difficult OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: modelling | exponential function

22. The population of a bacterial culture triples every hour. When the scientist observed the culture, it had already been growing for some time. She developed the equation for the population, P, after t hours as $P = 700(3)^t$, based on $t = 0$ representing the time she started her measurements. How many bacterial cells were there 2 h before she started measuring?

A 78
B 26
C 350
D 233

ANS: A DIF: Easy OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: negative exponents

23. Solve for x.

$1\,562\,500 = 4(5)^x$

A 9
B 7
C 8
D 11

ANS: C DIF: Easy OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: change of base

24. Solve for x, to one decimal place.

$7333 = 5^x$

A 1466.6
B 11.1
C 36 667.0
D 5.5

ANS: D DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: exponential equation | systematic trial

25. Solve for x.

$(36)^{3x} = 216^{(x+7)}$

A 0.3
B 7
C 6
D 3.0

ANS: B DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: exponential equation | equate exponents

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26. The half-life of a radioactive element can be modelled by $M = M_0\left(\frac{1}{32}\right)^{\frac{t}{45}}$, where M_0 is the initial mass of the element; t is the elapsed time, in hours; and M is the mass that remains after time t. The half-life of the element is

A 11 h
B 10 h
C 18 h
D 9 h

ANS: D DIF: Difficult OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: half-life | exponential decay

COMPLETION

1. The graph of $y = -(2)^{-x}$ has domain _____ and range _____.

ANS: $\{x|x \in \mathrm{R}\}$, $\{y|y < 0, y \in \mathrm{R}\}$

DIF: Average OBJ: Section 7.1 | Section 7.2 NAT: RF9
TOP: Characteristics of Exponential Functions | Transformations of Exponential Functions
KEY: domain | range | exponential function

2. The graph of $y = -(8)^{x}$ has a y-intercept of _____ and an x-intercept of _____.

ANS: -1, none

DIF: Average OBJ: Section 7.1 | Section 7.2 NAT: RF9
TOP: Characteristics of Exponential Functions | Transformations of Exponential Functions
KEY: exponential function | intercepts

3. The equation of the horizontal asymptote of the function $y = 4(8)^{-6(x-7)} - 9$ is _____.

ANS: $y = -9$

DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: asymptote | exponential function

4. The expression $\left(\sqrt{2^6}\right)\left(\sqrt[6]{4096}\right)$ written as a single power of 2 is _____.

ANS: 2^5

DIF: Easy OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: change of base

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5. The value of t that makes the equation $64^{5t} = 16^{t+1}$ true is _____.

ANS: $\frac{2}{13}$

DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: change of base

6. For the function $y = b^{kx}$, if $k < 0$, the graph of the function is _____ in the _____-axis, when compared to the graph of $y = b^x$.

ANS: reflected, y

DIF: Easy OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

MATCHING

Match the definition or explanation to its corresponding term.

A reflection in the y-axis
B exponential decay
C decreasing function
D $y = 0$
E vertical translation up or down

1. a pattern of growth in which each term is multiplied by a constant amount (between 0 and 1) to produce the next term

2. an exponential function of the form $y = a(c)^x$, where $a > 0$ and $0 < c < 1$

3. the horizontal asymptote of an exponential function of the form $y = a(c)^x$

4. occurs for the graph of $y = a(c)^{b(x-h)} + k$ when $k \neq 0$

5. occurs for the graph of $y = a(c)^{bx}$ when $b < 0$

1. ANS: B DIF: Easy OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: exponential growth | exponential decay

2. ANS: C DIF: Easy OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: increasing | decreasing | exponential function

3. ANS: D DIF: Easy OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: asymptote | exponential function

4. ANS: E DIF: Easy OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

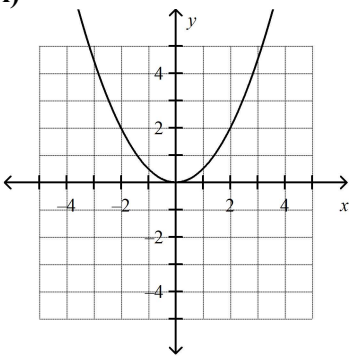
5. ANS: A DIF: Easy OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions

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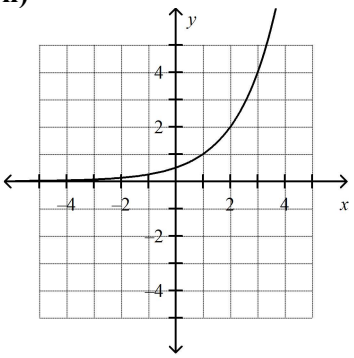
SHORT ANSWER

1. **a)** Determine the type of function shown in each graph.

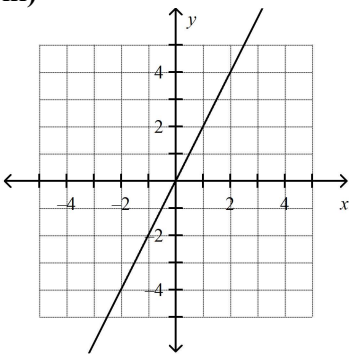
i)



ii)



iii)



b) Describe what you would expect to see in the first differences column of a table of values for each graph in part a).

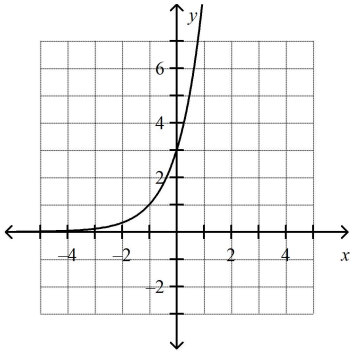
ANS:
a) i) quadratic
ii) exponential
iii) linear
b) i) successive values would be increasing by a constant amount
ii) successive values would be increasing by a constant factor
iii) all values would be constant

DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: linear | quadratic | exponential function

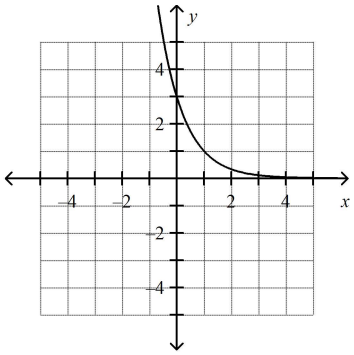
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2. Match each graph with the correct corresponding equation.

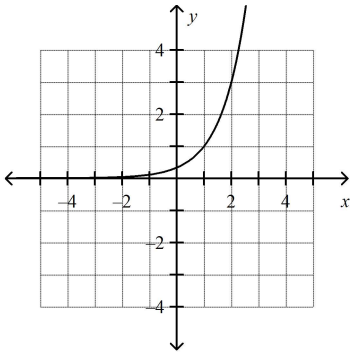
a)



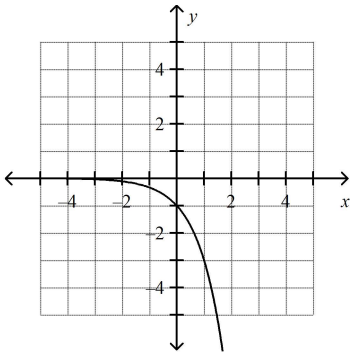
b)



c)



d)



i) $y = \frac{1}{3}(3)^x$

ii) $y = 3(3)^x$

iii) $y = -3^x$

iv) $y = 3\left(\frac{1}{3}\right)^x$

ANS:
a) ii)
b) iv)
c) i)
d) iii)

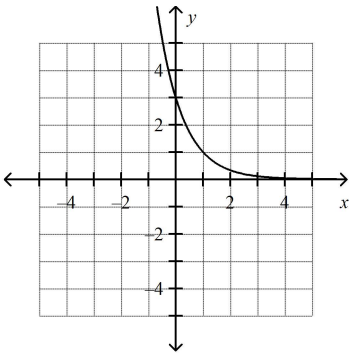
DIF: Easy OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: equation | graph | exponential function

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3. Sketch the graph of an exponential function with all of the following characteristics:
- domain $\{x | x \in \mathrm{R}\}$
- range $\{y | y > 0, y \in \mathrm{R}\}$
- y-intercept of 3
- no x-intercept
- the function is always decreasing

ANS:
Answers may vary.

Sample answer: The graph of $y = 3\left(\frac{1}{3}\right)^x$:



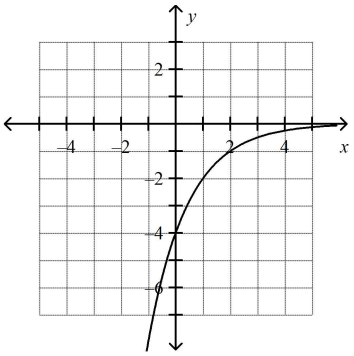
DIF: Average OBJ: Section 7.1 | Section 7.2 NAT: RF9
TOP: Characteristics of Exponential Functions | Transformations of Exponential Functions
KEY: domain | range | intercepts | exponential function

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4. Sketch an exponential function with all of the given conditions:
- domain $\{x | x \in \mathrm{R}\}$
- range $\{y | y < 0, y \in \mathrm{R}\}$
- y-intercept of -4
- no x-intercept
- the function is always increasing

ANS:
Answers may vary.

Sample answer: The graph of $y = -4\left(\frac{1}{2}\right)^x$:



DIF: Easy OBJ: Section 7.1 | Section 7.2 NAT: RF9
TOP: Characteristics of Exponential Functions | Transformations of Exponential Functions
KEY: domain | range | intercepts | exponential function

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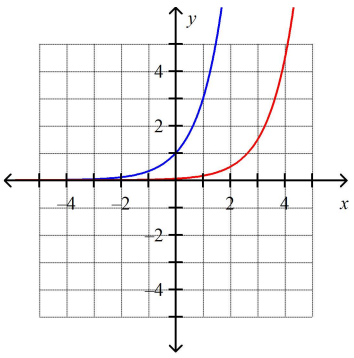
5. For the function $y = \frac{1}{2}(3)^{x-2}$,

a) describe the transformations of the function when compared to the function $y = 3^x$
b) sketch the graph of the given function and $y = 3^x$ on the same set of axes
c) state the domain, the range, and the equation of the asymptote

ANS:

a) a vertical compression by a factor of $\frac{1}{2}$ and a translation of 2 units to the right

b) The graph of $y = 3^x$ is shown in blue and the graph of $y = \frac{1}{2}(3)^{x-2}$ is shown in red.



c) domain $\{x | x \in \mathrm{R}\}$, range $\{y | y > 0, y \in \mathrm{R}\}$, $y = 0$

DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: graph | transformations of exponential functions

6. Write the equation for the function that results from each transformation or set of transformations applied to the base function $y = 5^x$.

a) reflect in the y-axis
b) shift 3 units to the right
c) shift 1 unit down and 4 units to the left
d) reflect in the x-axis and shift 2 units down

ANS:

a) $y = 5^{-x}$

b) $y = 5^{x-3}$

c) $y = 5^{x+4} - 1$

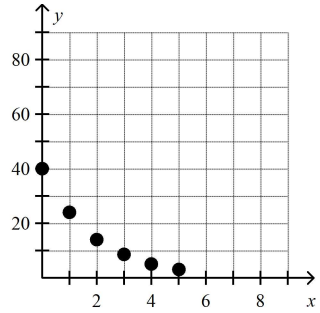
d) $y = -5^x - 2$

DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: equation | transformations of exponential functions

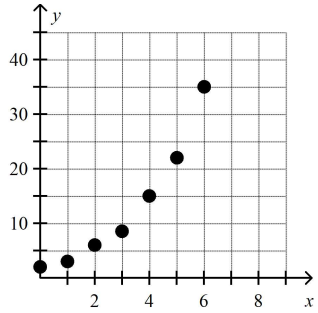
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7. Match each exponential scatter plot with the corresponding equation of its curve of best fit.

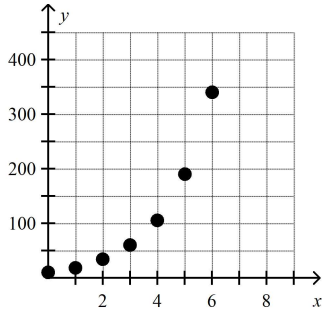
a)



b)



c)



i) $y = 2(1.6)^x$

ii) $y = 40(0.6)^x$

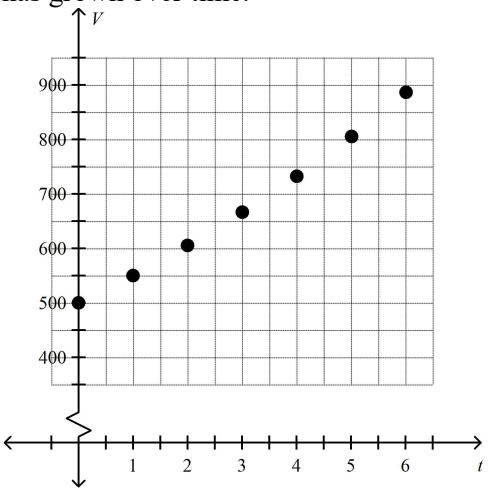
iii) $y = 10(1.8)^x$

ANS:
a) ii)
b) i)
c) iii)

DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: graph | modelling | exponential function

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8. Michael has invested some money with a financial institution. The scatter plot shows how his money has grown over time.



a) Do the data show an exponential trend? Explain your reasoning.
b) Estimate values of a and b to develop a model for the information of the form $V(t) = ab^t$, for t in years.
c) Use your model to predict the expected value of Michael's investment in 30 years.
d) How does the result in part c) illustrate the power of exponential growth?

ANS:
a) The data seem to show exponential growth, as there seems to be an increasing upward trend to the data.
b) The second data point is (1, 550), which implies a b-value of 1.1. Check this with the other data points.

t	$P = 500(1.1)^t$
0	500.00
1	550.00
2	605.00
3	665.50
4	732.05
5	805.26
6	885.78

These seem to match the graph. Thus, $a = 500$ and $b = 1.1$; $V(t) = 500(1.1)^t$.

c) $V(t) = 500(1.1)^t$

$V(30) = 500(1.1)^{30}$

≈ 8724.70

Michael's investment will be worth \$8724.70 in 30 years.
d) This calculation shows that as time increases, the value of the investment increases and the rate of change of the value increases at the same time.

DIF: Average OBJ: Section 7.1 | Section 7.3 NAT: RF9 | RF10
TOP: Characteristics of Exponential Functions | Solving Exponential Equations
KEY: modelling | exponential function

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9. Solve for n: $9^{n-1} = \left(\frac{1}{3}\right)^{4n-1}$

ANS:

$$9^{n-1} = \left(\frac{1}{3}\right)^{4n-1}$$

$$\left(3^2\right)^{n-1} = \left(3^{-1}\right)^{4n-1}$$

$$3^{2n-2} = 3^{1-4n}$$

Equate the exponents:

$$2n-2 = 1-4n$$

$$6n = 3$$

$$n = \frac{1}{2}$$

DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: change of base

10. Solve for x: $3^x = 9^{x^2-\frac{1}{2}}$

ANS:

$$3^x = 9^{x^2-\frac{1}{2}}$$

$$3^x = 3^{2\left(x^2-\frac{1}{2}\right)}$$

Equate the exponents:

$$x = 2x^2 - 1$$

$$0 = 2x^2 - x - 1$$

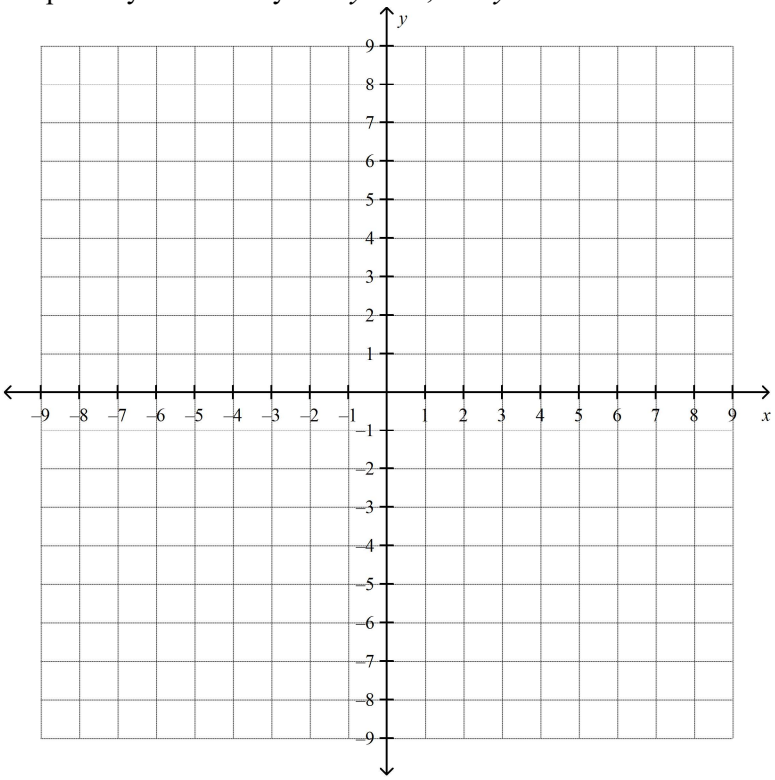
$$0 = (2x+1)(x-1)$$

$$x = -\frac{1}{2}, \quad x = 1$$

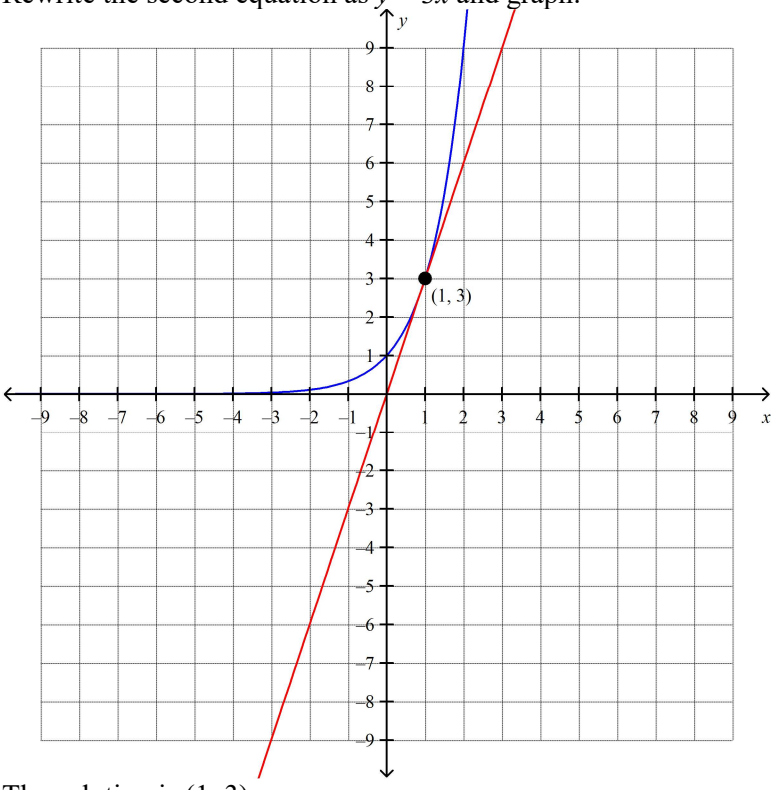
DIF: Difficult OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: change of base | equate exponents

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11. Graphically solve the system $y = 3^x$, $3x - y = 0$.



ANS:
Rewrite the second equation as $y = 3x$ and graph:



The solution is (1, 3).

DIF: Difficult + OBJ: Section 7.3 NAT: RF10

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TOP: Solving Exponential Equations KEY: system of equations | solve by graphing

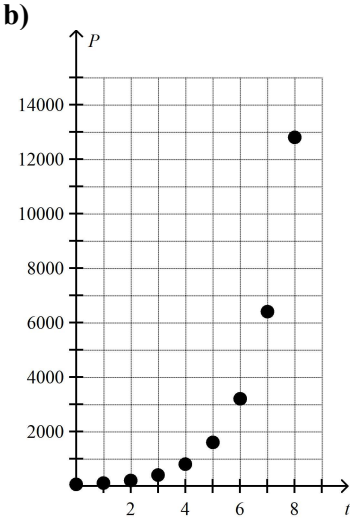
PROBLEM

1. A colony of ants starts with an initial population of 50 and doubles every week for 8 weeks.
a) Create a table of values for weeks 0 to 8 for the population of the colony.
b) Graph the data from your table of values.
c) Is the relationship between the ant population and the number of weeks exponential? Explain.
d) Model the information using an equation.

ANS:
a)

Time, t (weeks)	Population, P
0	50
1	100
2	200
3	400
4	800
5	1 600
6	3 200
7	6 400
8	12 800

b)



c) The data seem to be exponential, since the graph increases at an increasing rate. The values for population are being multiplied by a factor of 2 between successive terms in the table of values.
d) $P = 50(2)^t$

DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: graph | modelling | exponential growth

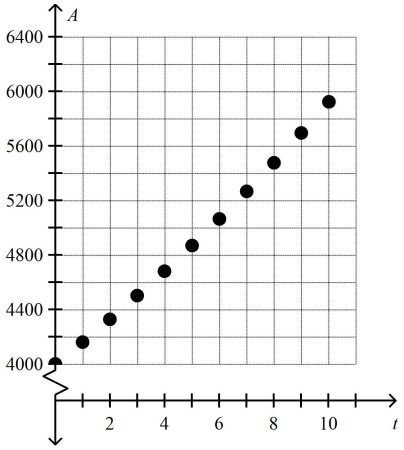
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2. An initial investment of $4000 earns interest at 4% per year, compounded annually.
a) Create a table of values for the first 10 years of the investment. Round the amounts to the nearest cent.
b) Graph the data from your table of values.
c) Is the relationship between the time and the amount of the investment exponential? Explain.
d) Write an equation to represent the data.

ANS:
a)

Time, t (years)	Amount, A ($)
0	4000.00
1	4160.00
2	4326.40
3	4499.46
4	4679.43
5	4866.61
6	5061.28
7	5263.73
8	5474.28
9	5693.25
10	5920.98

b)



c) The data seem to be exponential, since the graph increases at an increasing rate. The values for the amount are being multiplied by a factor of 1.04 between successive terms in the table of values.
d) $A = 4000(1.04)^t$

DIF: Average OBJ: Section 7.1 NAT: RF9
TOP: Characteristics of Exponential Functions
KEY: graph | modelling | exponential growth

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3. Jeff buys a new vehicle for $35 000. It is known that the vehicle will depreciate by 20% of its current value every year.

a) Write an equation to relate the depreciated value, V, of the vehicle to the age, t, in years, of the vehicle.

b) Use the equation to determine the value of the vehicle 2 years after Jeff buys it.

c) Approximately how long will it take the vehicle to depreciate to $3000?

ANS:

a) $V = 35\,000(0.80)^t$

b) $V = 35\,000(0.80)^t$

$= 35\,000(0.80)^2$

$= 22\,400$

The value of the vehicle after 2 years is $22 400.

c) $V = 35\,000(0.80)^t$

$3000 = 35\,000(0.80)^t$

Use systematic trial. When $t = 11$, $V = 3006.48$. Therefore, after approximately 11 years, the vehicle will be worth $3000.

DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: modelling | exponential decay | systematic trial

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4. Cobalt-60, which has a half-life of 5.3 years, is used in medical radiology. A sample of 60 mg of the material is present today.
a) Write an equation to relate the amount of cobalt-60 remaining and the number of half-life periods.
b) What amount will be present in 10.6 years?
c) How many years will it take for the amount of cobalt-60 to decay to 12.5% of its initial amount?

ANS:

a) $A = 60\left(\frac{1}{2}\right)^n$, where A is the amount of cobalt-60 remaining, in milligrams, and n is the number of half-life periods.

b) 10.6 years equals 2 half-life periods, since 5.3 × 2 = 10.6.

$$A = 60\left(\frac{1}{2}\right)^n$$

$$= 60\left(\frac{1}{2}\right)^2$$

$$= \frac{60}{4}$$

$$= 15$$

15 mg will be present in 10.6 years.

c) 12.5% = 0.125

$$= \frac{1}{8}$$

$$\frac{1}{8} = \left(\frac{1}{2}\right)^n$$

$$\left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^n$$

$$3 = n$$

It will take 5.3 × 3, or 15.9 years, for the amount of cobalt-60 to decay to 12.5% of its initial amount.

DIF: Difficult OBJ: Section 7.1 | Section 7.3 NAT: RF9 | RF10
TOP: Characteristics of Exponential Functions | Solving Exponential Equations
KEY: modelling | exponential decay | change of base

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5. Buket wants to invest some money in an account that pays 3.5% interest, compounded yearly. She wants to have an accumulated amount of $6000 at the end of 4 years.

a) Identify the values of A, i, and n to be used in the formula $P = A(1+i)^{-n}$.

b) Determine how much Buket needs to invest today in order for her to reach her goal in 4 years.

c) How much less would she need to invest if the financial institution were to offer her 4% annual interest? Show all your work.

ANS:

a) $A = 6000$, $i = 0.035$, $n = 4$

b) $P = A(1+i)^{-n}$

$= 6000(1.035)^{-4}$

≈ 5228.65

Therefore, she needs to invest $5228.65.

c) $P = A(1+i)^{-n}$

$= 6000(1.04)^{-4}$

≈ 5128.83

Therefore, if the financial institution were to offer 4% annual interest, she would be able to invest approximately $100 less to have the same accumulated amount at the end of 4 years.

DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: modelling | exponential decay | negative exponents

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6. When interest is compounded semi-annually, the formula used to find the amount of an investment is $A = P\left(1 + \frac{i}{2}\right)^{2n}$, where A represents the amount; P represents the principal invested; i represents the annual interest rate, as a decimal; and n represents the number of years of the investment.

a) Use the formula to determine the amount that each investment would be worth.

i) \$5000 at a rate of 4%, compounded semi-annually, for 10 years

ii) \$4000 at a rate of 5%, compounded semi-annually, for 20 years

b) If interest is compounded quarterly, the formula becomes $A = P\left(1 + \frac{i}{4}\right)^{4n}$. Use the formula to determine the amount that the investments from part a) would be worth if interest were compounded quarterly.

c) Explain the difference in the answers for parts a) and b).

ANS:

a) i) $A = P\left(1 + \frac{i}{2}\right)^{2n}$

$= 5000\left(1 + \frac{0.04}{2}\right)^{2 \times 10}$

$= 5000(1.02)^{20}$

≈ 7429.74

The investment would be worth \$7429.74.

ii) $A = P\left(1 + \frac{i}{2}\right)^{2n}$

$= 4000\left(1 + \frac{0.05}{2}\right)^{2 \times 20}$

$= 4000(1.025)^{40}$

$\approx 10\,740.26$

The investment would be worth \$10 740.26.

b) i) $A = P\left(1 + \frac{i}{4}\right)^{4n}$

$= 5000\left(1 + \frac{0.04}{4}\right)^{4 \times 10}$

$= 5000(1.01)^{40}$

≈ 7444.32

The investment would be worth \$7444.32.

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ii) $A = P\left(1 + \frac{i}{4}\right)^{4n}$

$= 4000\left(1 + \frac{0.05}{4}\right)^{4 \times 20}$

$= 4000(1.0125)^{80}$

$\approx 10\,805.94$

The investment would be worth \$10 805.94.

c) When the number of times interest is compounded per year is increased, the value at the end of the investment period increases.

DIF: Difficult OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: evaluate exponential functions

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7. A radioactive sample with an initial mass of 72 mg has a half-life of 10 days.
a) Write a function to relate the amount remaining, A, in milligrams, to the time, t, in days.
b) What amount of the radioactive sample will remain after 20 days?
c) What amount of the radioactive sample was there 30 days ago?
d) How long, to the nearest day, will it take for there to be 0.07 mg of the initial sample remaining?

ANS:

a) $A = 72\left(\frac{1}{2}\right)^{\frac{t}{10}}$

b) $A = 72\left(\frac{1}{2}\right)^{\frac{t}{10}}$

$= 72\left(\frac{1}{2}\right)^{\frac{20}{10}}$

$= 72\left(\frac{1}{2}\right)^{2}$

$= 18$

There will be 18 mg remaining after 20 days.

c) $A = 72\left(\frac{1}{2}\right)^{\frac{t}{10}}$

$= 72\left(\frac{1}{2}\right)^{\frac{-30}{10}}$

$= 72\left(\frac{1}{2}\right)^{-3}$

$= 576$

There was 576 mg 30 days ago.

d) $A = 72\left(\frac{1}{2}\right)^{\frac{t}{10}}$

$0.07 = 72\left(\frac{1}{2}\right)^{\frac{t}{10}}$

$\frac{0.07}{72} = \left(\frac{1}{2}\right)^{\frac{t}{10}}$

Use systematic trial.

$\frac{0.07}{72} \doteq 0.000\,972$

For $t = 100$, $\left(\frac{1}{2}\right)^{10} \doteq 0.000\,977$.

It will take approximately 100 days for there to be 0.07 mg remaining.

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DIF: Average OBJ: Section 7.2 | Section 7.3 NAT: RF9 | RF10
TOP: Transformations of Exponential Functions | Solving Exponential Equations
KEY: modelling | evaluate exponential functions

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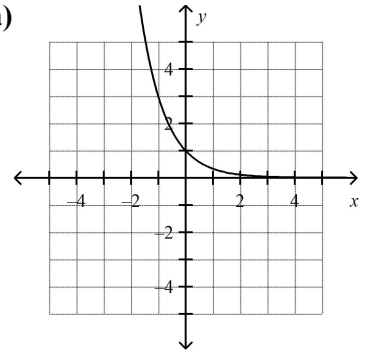
8. Graph each function. Identify the
i) domain
ii) range
iii) intercepts
iv) intervals of increase/decrease
v) equation of the asymptote

a) $f(x) = \left(\frac{1}{3}\right)^x$

b) $g(x) = -(2)^{-x}$

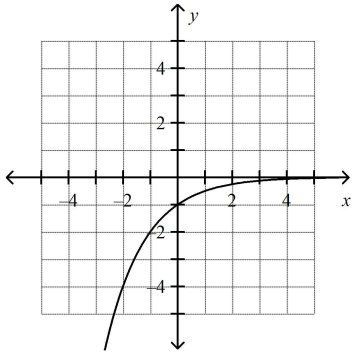
ANS:

a)



i) domain $\{x | x \in \mathrm{R}\}$
ii) range $\{y | y > 0, y \in \mathrm{R}\}$
iii) no x-intercept; y-intercept is 1
iv) the function is always decreasing
v) $y = 0$

b)



i) domain $\{x | x \in \mathrm{R}\}$
ii) range $\{y | y < 0, y \in \mathrm{R}\}$
iii) no x-intercept; y-intercept is –1
iv) the function is always increasing
v) $y = 0$

DIF: Average OBJ: Section 7.1 | Section 7.2 NAT: RF9
TOP: Characteristics of Exponential Functions | Transformations of Exponential Functions
KEY: graph | domain | range | intercepts | asymptote | exponential function

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9. **a)** Rewrite the function $y = 2^{-2x+4} + 6$ in the form $y = a(2)^{b(x-h)} + k$.

b) Describe the transformations that must be applied to the graph of $y = 2^x$ to obtain the graph of the given function.

c) Graph the function.

d) Determine the equation of the function that results after the graph in part c) is reflected in the x-axis.

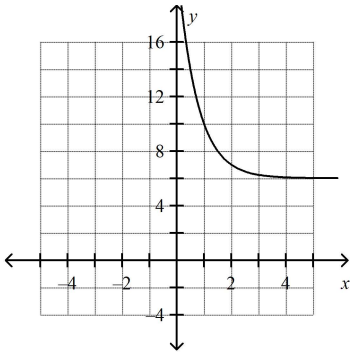
e) Graph the function from part d).

ANS:

a) $y = 2^{-2(x-2)} + 6$

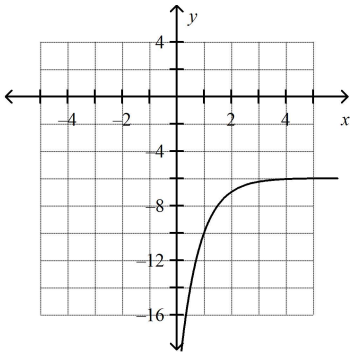
b) Reflect in the y-axis, compress horizontally by a factor of $\frac{1}{2}$, and translate 2 units to the right and 6 units up.

c)



d) $y = -2^{-2(x-2)} - 6$

e)



DIF: Average OBJ: Section 7.2 NAT: RF9 TOP: Exponential Functions
KEY: graph | transformations of exponential functions

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10. **a)** Write the equation for a transformed exponential function with a base of 4 that passes through the point $\left(0, \frac{3}{2}\right)$.

b) Write two equations, different from the one in part a), that satisfy these criteria.
c) Use algebraic and/or graphical reasoning to explain why each equation is a solution.

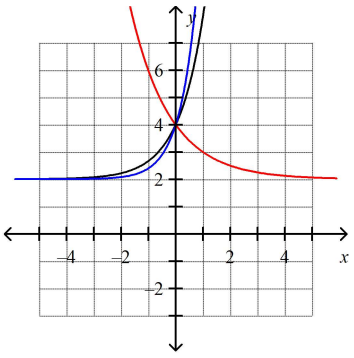
ANS:
Answers may vary. Sample answers:
a) $y = \frac{1}{2}(4)^{2x} + 1$

b) $y = 4^x + 0.5$; $y = -4^x + 2.5$
c) Either graph the equations to show that they all contain the given point or substitute $x = 0$ into each function to show that the result is the value $y = \frac{3}{2}$.

DIF: Difficult OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: transformations of exponential functions | equation

11. **a)** Write the equation for a transformed exponential function whose horizontal asymptote is $y = 2$, with a y-intercept of 4.
b) Is this function unique? Explain.
c) Illustrate your answer to part b) graphically, using examples where appropriate.

ANS:
Answers may vary. Sample answers:
a) $y = 2(3)^x + 2$
b) This function is not unique. Many exponential functions share these two properties, including functions with a different base.
c) The graph shows that the functions $y = 2(3)^x + 2$ (in black), $y = 2(5)^x + 2$ (in blue), and $y = 2\left(\frac{1}{2}\right)^x + 2$ (in red) all share the given properties.



DIF: Average OBJ: Section 7.2 NAT: RF9
TOP: Transformations of Exponential Functions
KEY: exponential function | transformations | equation

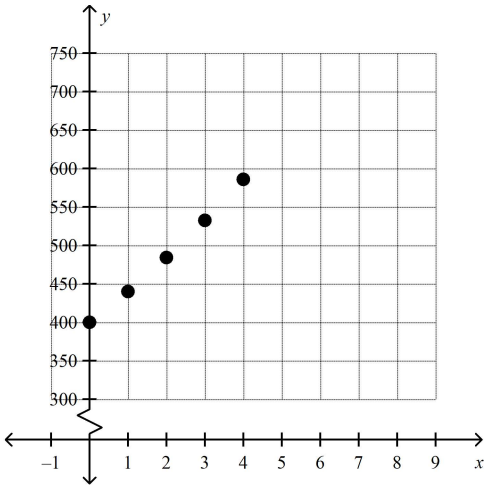
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12. An antique table was worth \$400 in 2000. Each subsequent year, the table increased in value until it was sold at an auction in 2004. The table shows the value of the antique table for the years 2000 to 2004.

Year	Value (\$)
2000	400.00
2001	440.00
2002	484.00
2003	532.40
2004	585.64

a) Create a scatter plot of the data.
b) Describe the shape of the curve.
c) Determine the equation for the curve of best fit, using $t = 0$ for the year 2000.
d) Use this equation to determine the value of the antique table in 2010, assuming that the value continues to increase according to the equation of the curve of best fit.
e) Approximately how long will it take for the table to be worth \$1250?

ANS:
a)



b) The data appear to be exponential.
c) The second data point is (1, 440), so the value of a is 1.1. The value at $t = 0$ is $V = 400$, so $V = 400(1.1)^t$.
Check that this matches the data in the table.

t	$V = 400(1.1)^t$
2000	400.00
2001	440.00
2002	484.00
2003	532.40
2004	585.64

Thus, $V = 400(1.1)^t$.
d) Using $t = 10$ for the year 2010,

$$V = 400(1.1)^t$$
$$= 400(1.1)^{10}$$
$$\approx 1037.50$$

In 2010, the value of the table will be \$1037.50.

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e) $V = 400(1.1)^t$

$1250 = 400(1.1)^t$

$3.125 = (1.1)^t$

With systematic trial, this occurs approximately when t = 12 years.

DIF: Difficult + OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: modelling | exponential function

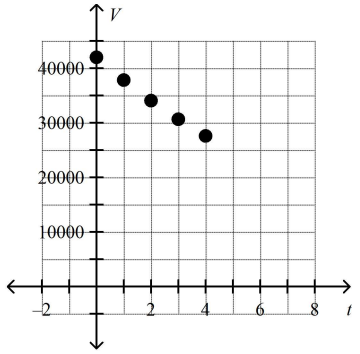
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13. The table shows the value, V, of Rachel's car t years after it was purchased.

Years After Purchase	Value ($)
0	42 000
1	37 800
2	34 020
3	30 618
4	27 556

a) Make a scatter plot for the data.
b) Describe the shape of the curve.
c) Determine an equation to model the data.
d) Use the equation to determine the value of the car in 8 years.
e) Approximately how long will it take for the car to be worth less than $10 000?

ANS:
a)



b) The data appear to be exponential.
c) From the table, the factor appears to be 0.9. Thus, $V = 42\,000(0.9)^t$.
d) Substitute $t = 8$.

$$V = 42\,000(0.9)^t$$
$$= 42\,000(0.9)^8$$
$$\approx 18\,079.62$$

The car will be worth approximately $18 079.62 in 8 years.
e) Substitute $V = 10\,000$ and solve for t.

$$V = 42\,000(0.9)^t$$
$$10\,000 = 42\,000(0.9)^t$$
$$0.238 \approx (0.9)^t$$

Using systematic trial, this occurs when t is approximately 14 years.

DIF: Difficult + OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: modelling | exponential function

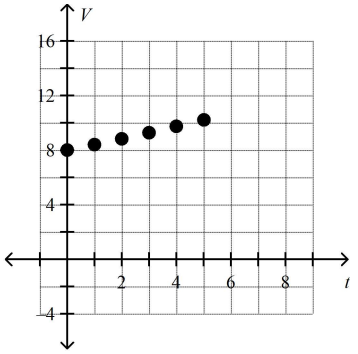
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14. Ranier has been researching the share price, V, of a stock for t weeks. The data are shown below.

Week	Price ($)
0	8.00
1	8.40
2	8.82
3	9.26
4	9.77
5	10.21

a) Make a scatter plot for the data.
b) Describe the shape of the curve.
c) Determine an equation to model the data.
d) Use the equation to determine the value of the stock in 9 weeks.
e) Approximately how long will it take for the share price of the stock to exceed $20?

ANS:
a)



b) The data appear to be exponential.
c) From the table, the factor appears to be 1.05. Thus, $V = 8(1.05)^t$.
d) Substitute $t = 9$.

$V = 8(1.05)^t$

$= 8(1.05)^9$

≈ 12.41

In 9 weeks, the stock will be worth approximately $12.41 per share.
e) Substitute $V = 20$ and solve for t.

$V = 8(1.05)^t$

$20 = 8(1.05)^t$

$2.5 = 1.05^t$

Using systematic trial, this occurs when t is approximately 19 weeks.

DIF: Difficult + OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: modelling | exponential function

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15. Solve the equation $\sqrt[3]{256^2} \times 16^x = 64^{x-3}$.

ANS:

$\sqrt[3]{256^2} \times 16^x = 64^{x-3}$

$(2^8)^{\frac{2}{3}} \times 2^{4x} = 2^{6x-18}$

$2^{4x+\frac{16}{3}} = 2^{6x-18}$

$4x + \frac{16}{3} = 6x - 18$

$-2x = -\frac{70}{3}$

$x = \frac{35}{3}$

DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: exponential equation | change of base

16. Solve the equation $2^{3x} = 4$.

ANS:

$2^{3x} = 2^2$

$3x = 2$

$x = \frac{2}{3}$

DIF: Easy OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: exponential equation | change of base

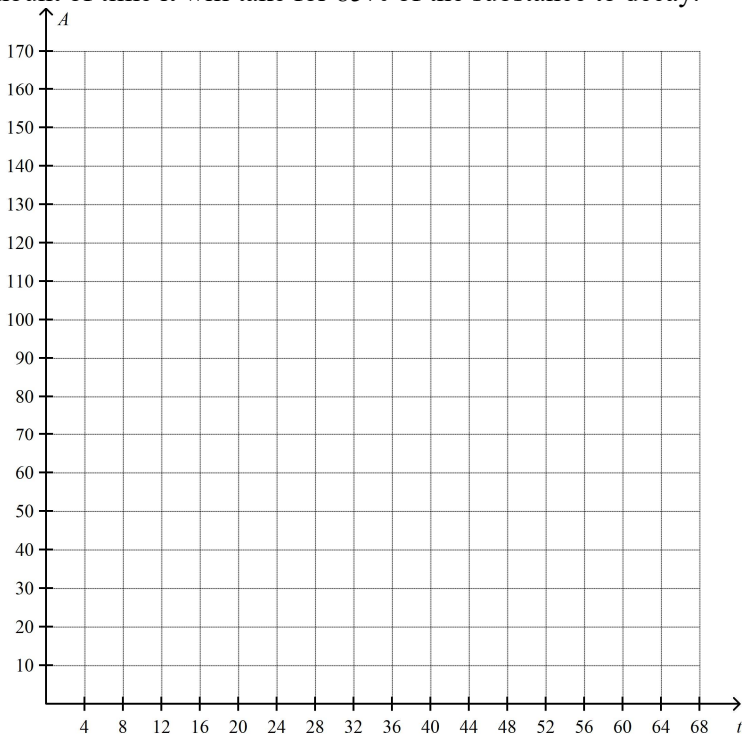
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17. A 200-g sample of a radioactive substance is placed in a chamber to be tested. After 3 h, 140 g of the sample remains.

a) Determine the half-life of this substance.

b) Determine the equation that represents the decay of the substance from your answer to part a).

c) Graph the amount of the substance remaining as a function of time. Use your graph to determine the amount of time it will take for 85% of the substance to decay.



ANS:

a) Let h represent the half-life.

$$A = 200\left(\frac{1}{2}\right)^{\frac{t}{h}}$$

$$140 = 200\left(\frac{1}{2}\right)^{\frac{3}{h}}$$

$$0.7 = \left(\frac{1}{2}\right)^{\frac{3}{h}}$$

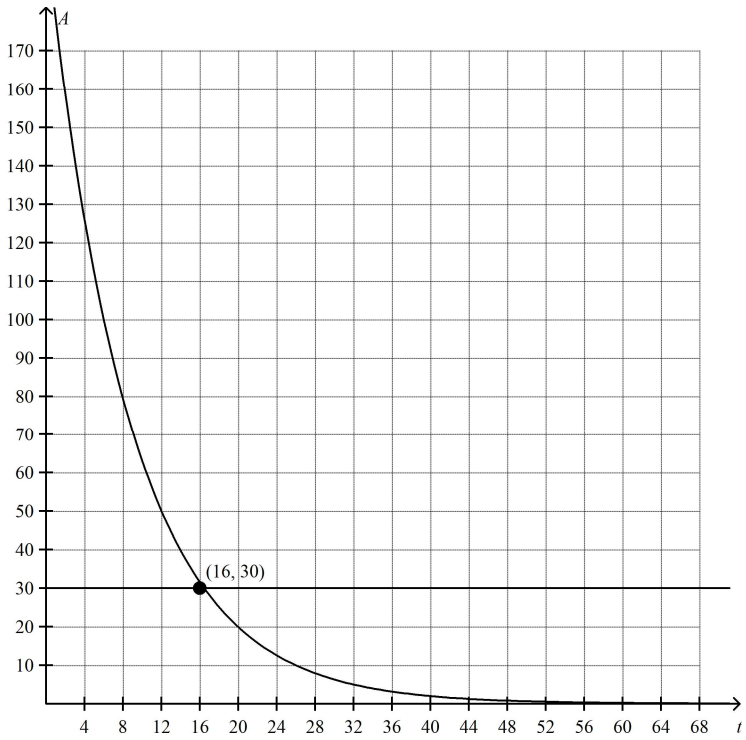
Use systematic trial to determine that when $\frac{3}{h} \approx 0.5$, the equation is true. From this, $h = 6$.

The half-life is 6 h.

b) $A = 200\left(\frac{1}{2}\right)^{\frac{t}{6}}$

c) Since 85% has decayed, then 15% or 30 g remains.

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From the graph, it would take approximately 16 h for the sample to decay to this point.

DIF: Difficult OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: exponential equation | half-life | graph

18. A \$21 500 investment earns 5.25% interest, compounded quarterly.
a) Determine the value of the investment in 5 years.
b) How long will it take the original investment to double in value?

ANS:

a) The quarterly interest rate is $\frac{5.25}{4}$ or 1.3125, and the number of compounding periods is 4×5 or 20.

$A = P(1+i)^n$

$= 21\,500(1+0.013\,125)^{20}$

$\approx 27\,906.10$

The value of the investment after 5 years is \$27 906.10.

b) $43\,000 = 21\,500(1+0.013125)^n$

$2 = (1.013125)^n$

Using systematic trial, n = 53.2 quarters or approximately 13.3 years.

DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: exponential equation | compound interest formula

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19. A chemist has a 20-mg sample of polonium-218. He needs approximately 81.5% of it for an experiment. Given that the half-life of polonium-218 is approximately 3.1 min, how many seconds will it take for the sample to decay to the desired mass?

ANS:
81.5% of 20 mg is 16.3 mg, and 3.1 min is 186 s.

$$16.3 = 20\left(\frac{1}{2}\right)^{\frac{t}{186}}$$

$$0.815 = 0.5^{\frac{t}{186}}$$

Using systematic trial, $\frac{t}{186} = 0.3$. Therefore, the time is approximately 56 s.

DIF: Average OBJ: Section 7.3 NAT: RF10
TOP: Solving Exponential Equations KEY: exponential equation | half-life

20. An investment offers a bonus of 2% of the principal after being invested for 5 years. If \$50 000 is invested at 4.75%, compounded annually, for 10 years, describe how the graph of the investment with the bonus differs from the graph of the investment without the bonus.

ANS:

same shape, vertical translation up by \$1000

DIF: Easy OBJ: Section 7.2 NAT: RF10
TOP: Characteristics of Exponential Functions | Transformations of Exponential Functions
KEY: exponential equation | compound interest formula | transformations

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